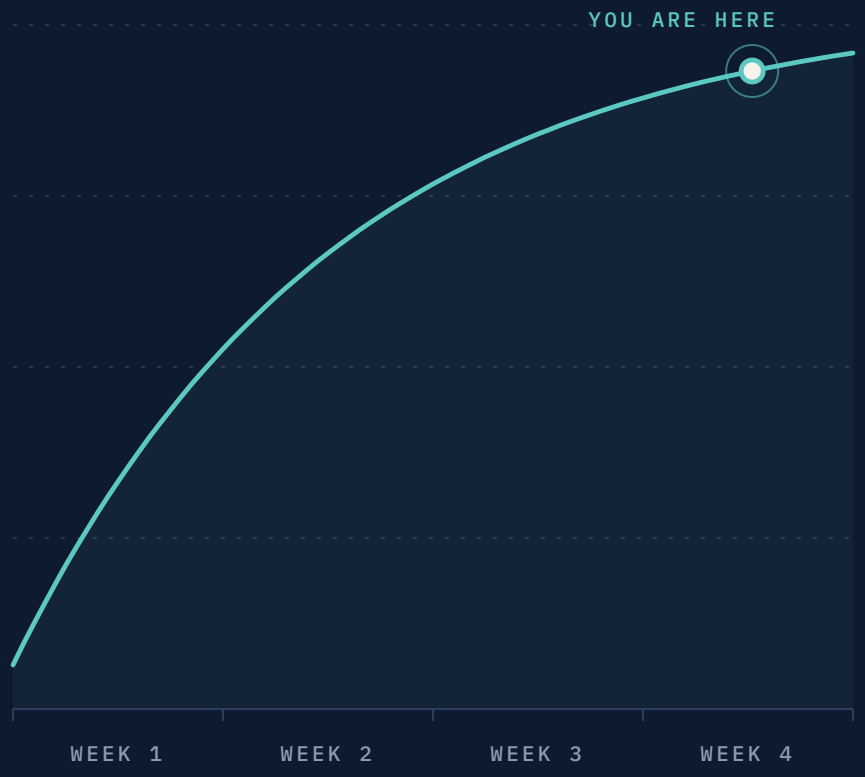
WEEK 4 · DAY 2, PART 2 · OVERVIEW

Analyzing Text, Part 2



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 8 SECTIONS / 5 EXERCISES

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The Desk Lead's Note

Part 1 proved the tools on one statement. Now my table: one row for each of the 46. How long each is, the rate range it states, how often a few plain words appear, and how much of its wording it shares with the statement before. Clean the text first so Inflation and inflation. count as one word, and show me what each step does to the count. Counts only: no reading of policy.

Your Route Through Day 2, Part 2

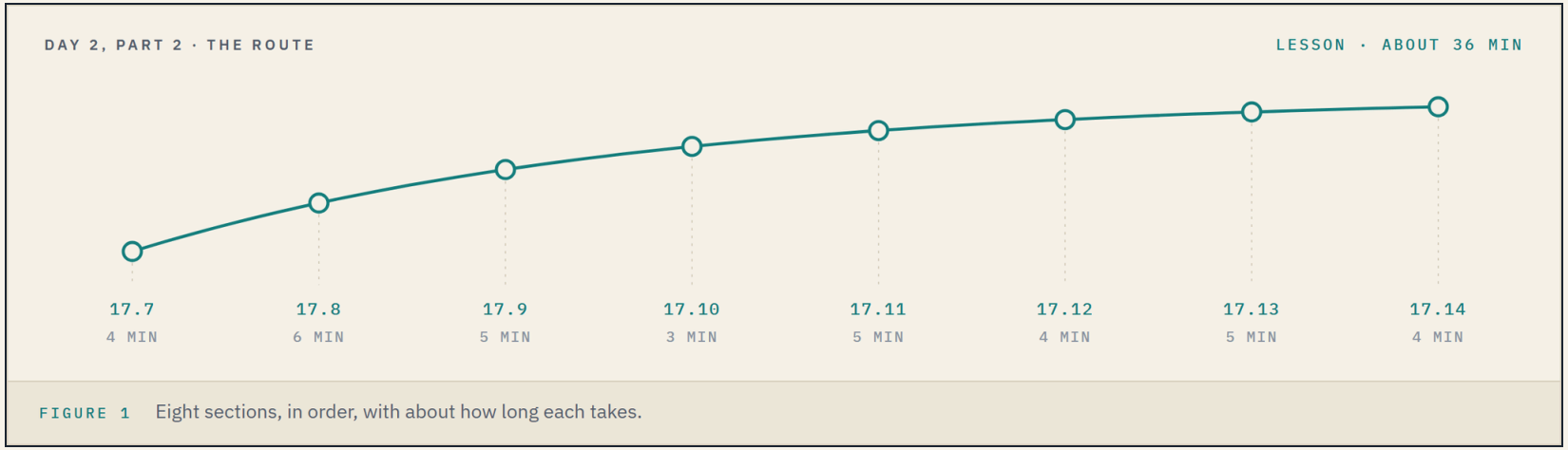


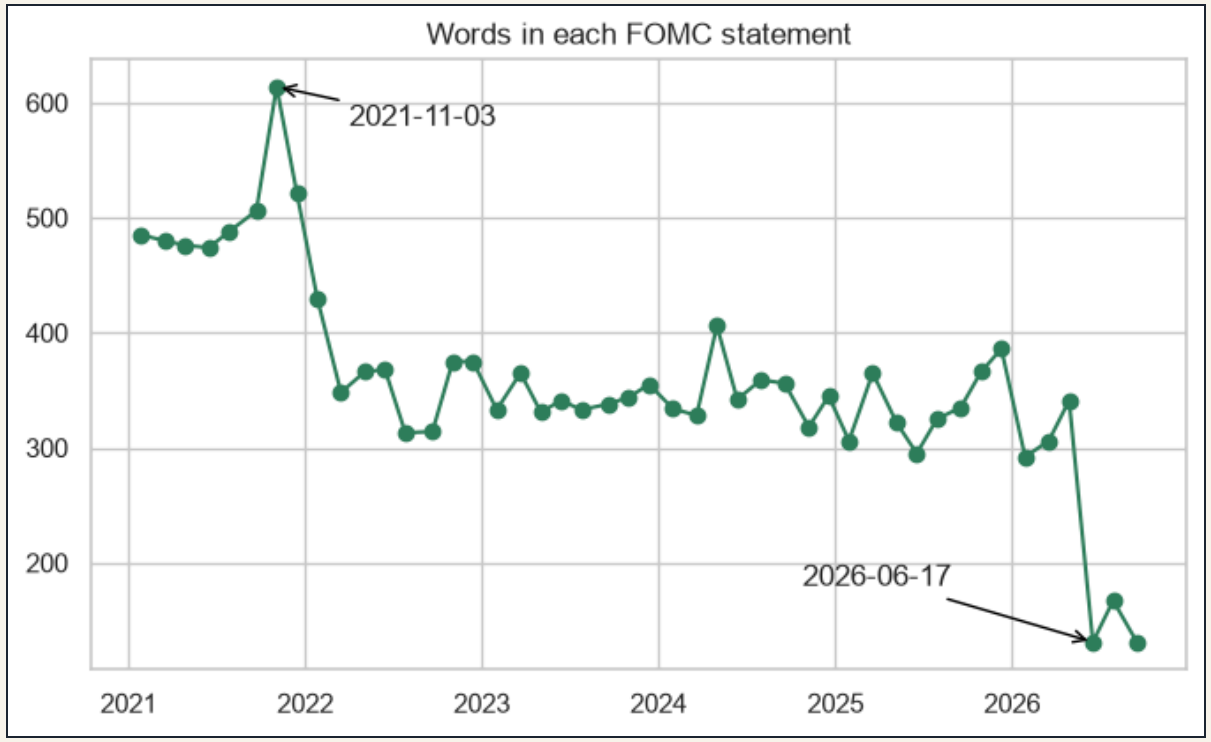
FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
17.7	Every Statement's Length	4 min
17.8	Cleaning Text in Steps	6 min
17.9	Stopwords	5 min
17.10	Stemming	3 min

	SECTION	ABOUT
17.11	The Range in Every Statement	5 min
17.12	Every Statement Cleaned and Counted	4 min
17.13	A Document-Term Table	5 min
17.14	How Much Wording Changed	4 min

Time. About 36 minutes for the lesson, then about 15 for the five exercises.

17.7 Every Statement's Length



LENGTH OF EACH STATEMENT

```
statements['words'] = statements['text'].str.split().str.len()

359.3 344.5
```

EXCEL SIDE BY SIDE

With one statement per row, `=LOWER(C2)` and `=LEN(C2)` filled down do what `.str.lower()` and `.str.len()` do.

Length needed no cleaning. Counting a word does.

17.8 Cleaning Text in Steps



FIGURE 2 Seven pieces of the range sentence through four cleaning steps: 23 pieces become 21 tokens.

FOUR STEPS IN ONE FUNCTION

```
def clean(text):
    text = re.sub(r'[^a-z\s]', ' ', text.lower())
    text = re.sub(r'\s+', ' ', text).strip()
    return text.split(' ')
print(clean(sentence) == tokens)
```

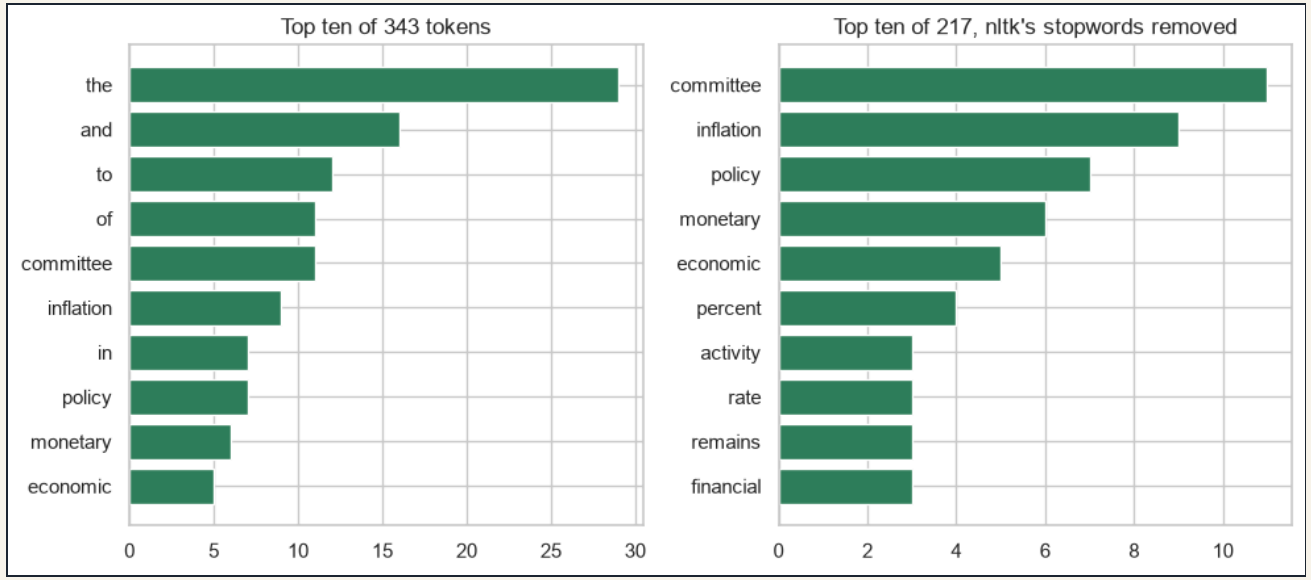
```
True
```

STEPS 1 AND 2

```
step1 = sentence.lower()
step2 = re.sub(r'[^a-z\s]', ' ', step1)
```

343 tokens, and more than a third of them say nothing.

17.9 Stopwords



REMOVE THE STOPWORDS

```
kept = [token for token in statement_tokens if token not in stop_words]
```

```
217 180 152
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Count one token	`counts['the']`	`=COUNTIF(B2:B344, "the")`
Is it a stopword	`token in stop_words`	`=COUNTIF(stop!A:A, B2)>0`, filled down
Keep the rest	`kept`, a comprehension	`=FILTER(B2:B344, COUNTIF(stop!A:A, B2:B344)=0)`

217 left, and `committee` leads. Forms of one word still count apart.

17.10 Stemming

ONE ROOT FOR SEVERAL FORMS

```
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
for word in ['remain', 'remained', 'remains', 'assess', 'assessing', 'assessments',
             'inflation', 'policy', 'committee', 'monetary', 'securities']:
    print(word, '->', stemmer.stem(word))
```

```
remain -> remain
remained -> remain
remains -> remain
assess -> assess
assessing -> assess
assessments -> assess
inflation -> inflat
policy -> polici
committee -> committe
monetary -> monetari
securities -> secur
```

STEMS IN THE STATEMENT

```
stems = [stemmer.stem(token) for token in kept]
print(len(stems), len(set(kept)), len(set(stems)))
```

```
217 152 146
```

EXCEL SIDE BY SIDE

No function stems; `=COUNTIF(B2:B344, "remain*")` counts the tokens that start `remain`: 5.

A stem count is rougher, so tokens it is. Now all 46, starting with the range.

17.11 The Range in Every Statement

HOW COMMON THE HIDDEN HYPHEN IS

```
print(statements['text'].str.contains('\u2011').sum())
print(statements['text'].str.count('\u2011').sum())
```

```
26
38
```

WIDEN THE CLASS

```
wide_pattern = r'[\d/\u2011-]+ to [\d/\u2011-]+ percent'
print(found_wide.str.len().value_counts().sort_index())
```

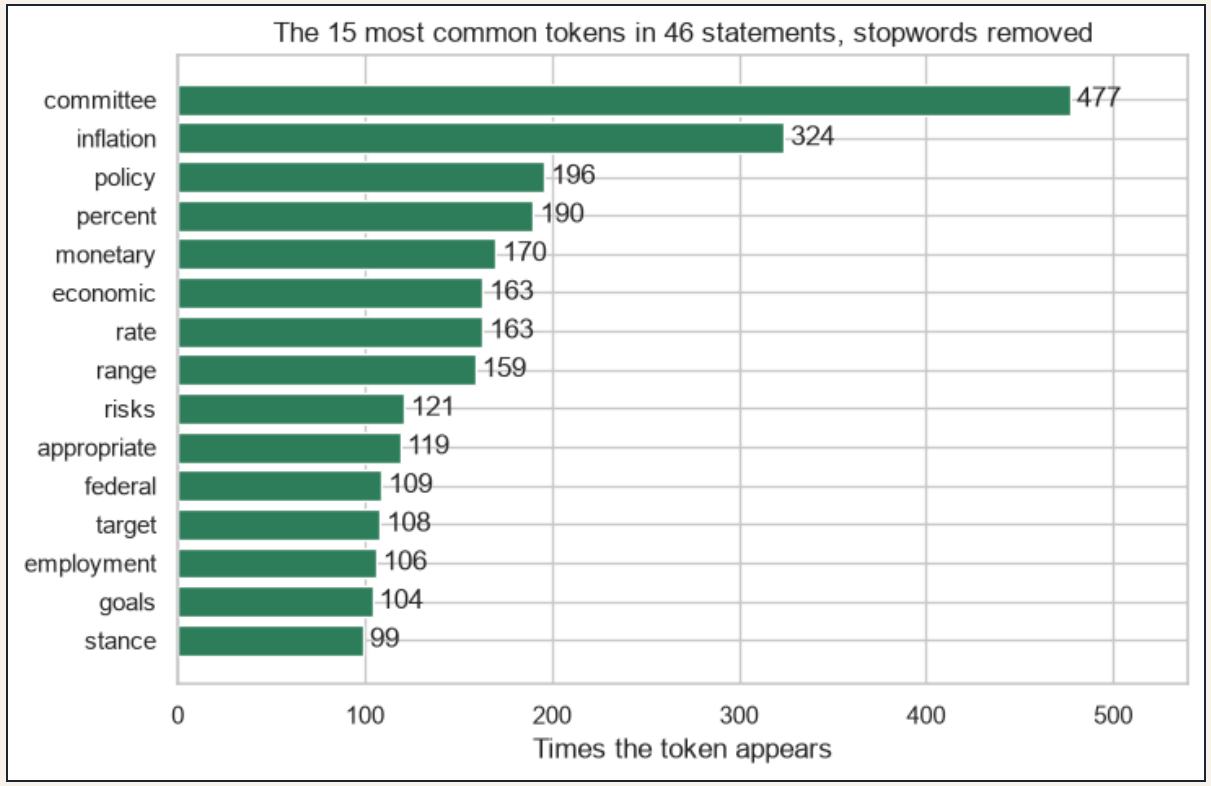
```
text
1    44
2     2
Name: count, dtype: int64
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
A character's code	`ascii()` writes `\u2011`	`=UNICODE(MID(C2, 715, 1))`
Swap it for a plain hyphen	`text.replace('\u2011', '-')`	`=SUBSTITUTE(C2, UNICHAR(8209), "-")`

46 of 46, and one row was wrong while the count looked right.

17.12 Every Statement Cleaned and Counted



CLEAN AND COUNT EVERY ROW

```
statements['n_tokens'] = statements['tokens'].str.len()
statements['n_kept'] = statements['kept'].str.len()
print(statements['n_tokens'].sum(), statements['n_kept'].sum())
```

```
16446 10204
```

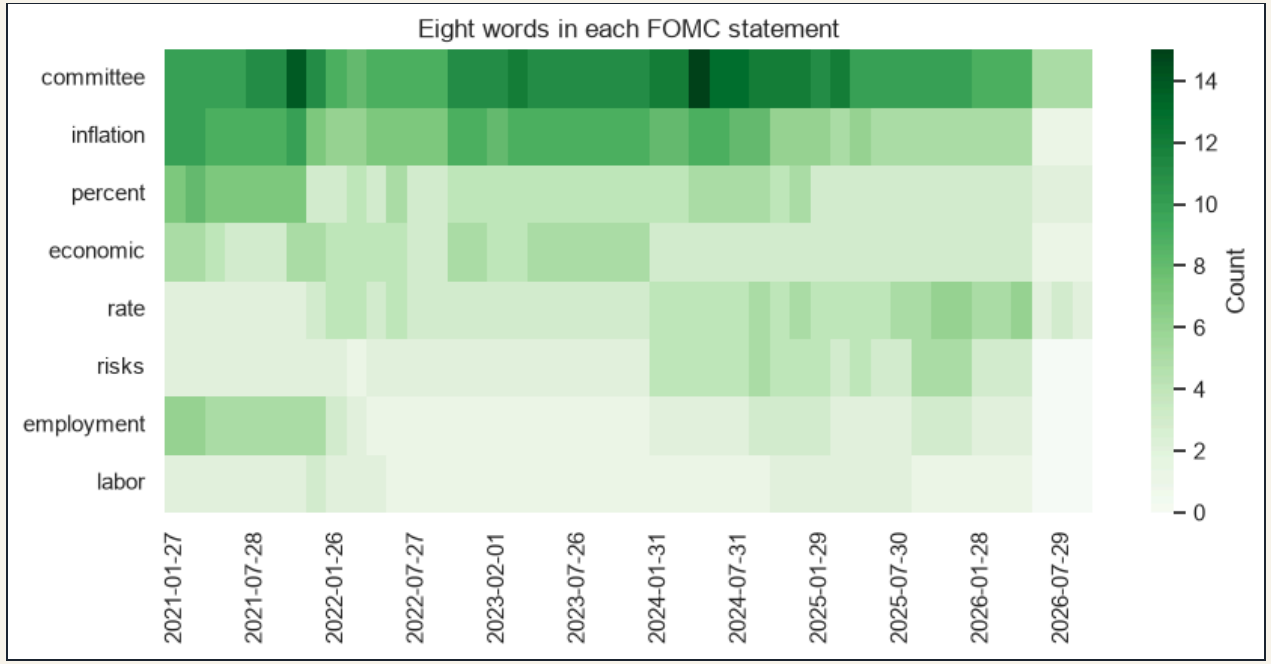
THE WHOLE FILE'S TOP TOKENS

```
print(len(all_counts), all_counts.most_common(3))
```

```
507 [('committee', 477), ('inflation', 324), ('policy', 196)]
```

10,204 tokens, and the row for 2023-11-01 ties. Next, the counts as a table.

17.13 A Document-Term Table



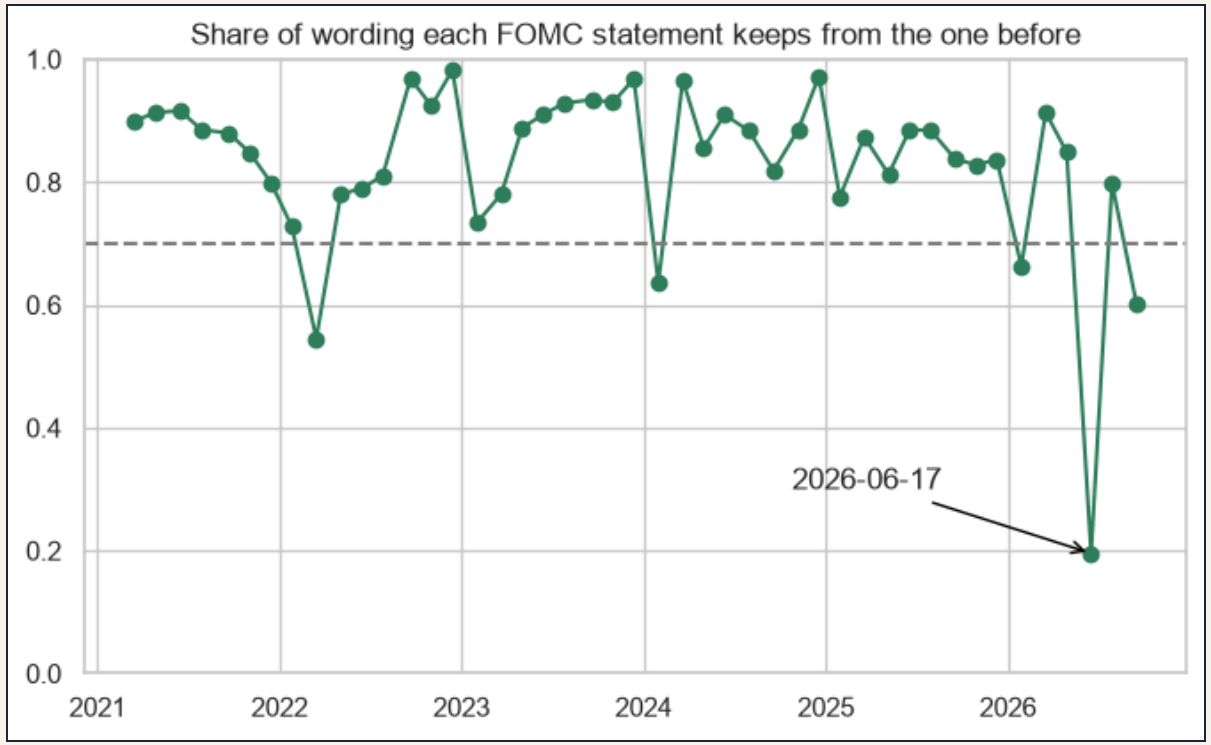
EIGHT WORDS, 46 ROWS

```
terms = ['committee', 'inflation', 'percent', 'economic',
         'rate', 'risks', 'employment', 'labor']
print(term_table.iloc[0].tolist())
print(term_table.sum().tolist())
```

```
[10, 10, 7, 5, 2, 2, 6, 2]
[477, 324, 190, 163, 163, 121, 106, 62]
```

A count leans on length; per 100 tokens fixes it, and a share counts each statement once. Last: how much each statement changed.

17.14 How Much Wording Changed



THE SCORE

```
def similarity(first, second):
    both = set(first) & set(second)
    either = set(first) | set(second)
    return len(both) / len(either)
```

AGAINST THE ONE BEFORE

```
statements['similarity'] = values
```

```
0.832 2026-06-17 0.196 2022-12-14 0.982
```

The note's four columns, every cell a count, a figure from the text, or a ratio of counts: no reading of policy.

You Can Now

- Clean text in four steps and keep them in one function.
- Count tokens with **`Counter`** and remove nltk's stopwords.
- Stem tokens, knowing the stemmer's two limits.
- Find a hidden character with **`ascii()`** and widen a pattern to take it.
- Run text methods on a column with **`.str`** and **`.apply()`**.
- Build a document-term table, scale it by length, and compare statements by shared words.

THE NOTE ASKED	ANSWER
How long each is	131 to 614 words
The range it states	46 of 46
A few plain words	10,204 kept tokens; 2023-11-01 ties at 343 and 217
Wording shared	a mean of 0.832, a low of 0.196

NEXT
WEEK 4 · DAY 3

Case Study: Statement Sentiment

Three word lists score 46 published statements; the scores meet each other, then yields.